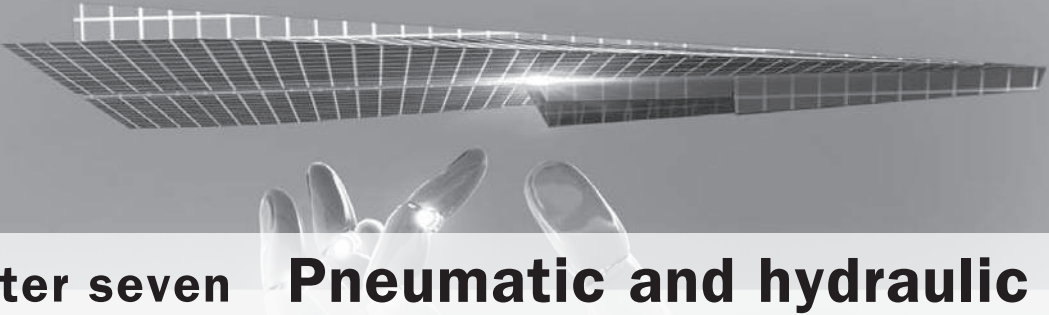




# Chapter seven **Pneumatic and hydraulic actuation systems**

## Objectives

The objectives of this chapter are that, after studying it, the reader should be able to:

- Interpret system drawings, and design simple systems, for sequential control systems involving hydraulic/pneumatic directional control valves and cylinders.
- Explain the principles of process control valves, their characteristics and sizing.

### 7.1

#### Actuation systems

Actuation systems are the elements of control systems which are responsible for transforming the output of a microprocessor or control system into a controlling action on a machine or device. Thus, for example, we might have an electrical output from the controller which has to be transformed into a linear motion to move a load. Another example might be where an electrical output from the controller has to be transformed into an action which controls the amount of liquid passing along a pipe.

In this chapter fluid power systems, namely pneumatic and hydraulic actuation systems, are discussed. **Pneumatics** is the term used when compressed air is used and **hydraulics** when a liquid, typically oil. In Chapter 8 mechanical actuator systems are discussed and in Chapter 9 electrical actuation systems.

### 7.2

#### Pneumatic and hydraulic systems

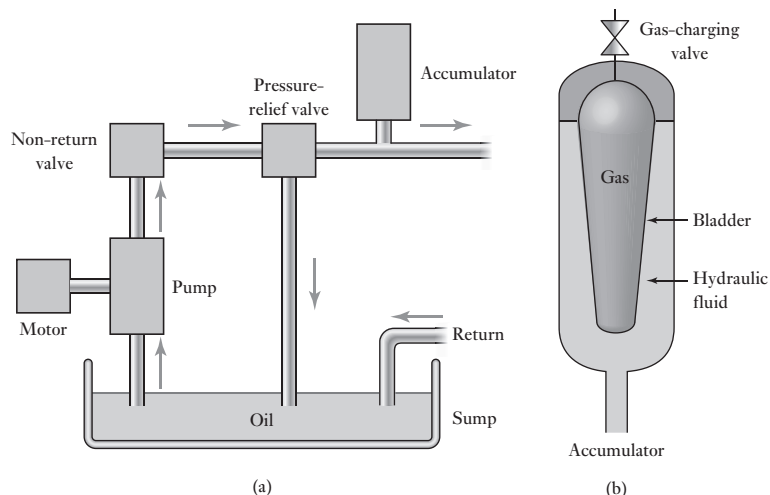
Pneumatic signals are often used to control final control elements, even when the control system is otherwise electrical. This is because such signals can be used to actuate large valves and other high-power control devices and so move significant loads. The main drawback with pneumatic systems is, however, the compressibility of air. Hydraulic systems can be used for even higher power control devices but are more expensive than pneumatic systems and there are hazards associated with oil leaks which do not occur with air leaks.

The atmospheric pressure varies with both location and time but in pneumatics is generally taken to be  $10^5$  Pa, such a pressure being termed 1 bar.

##### 7.2.1 Hydraulic systems

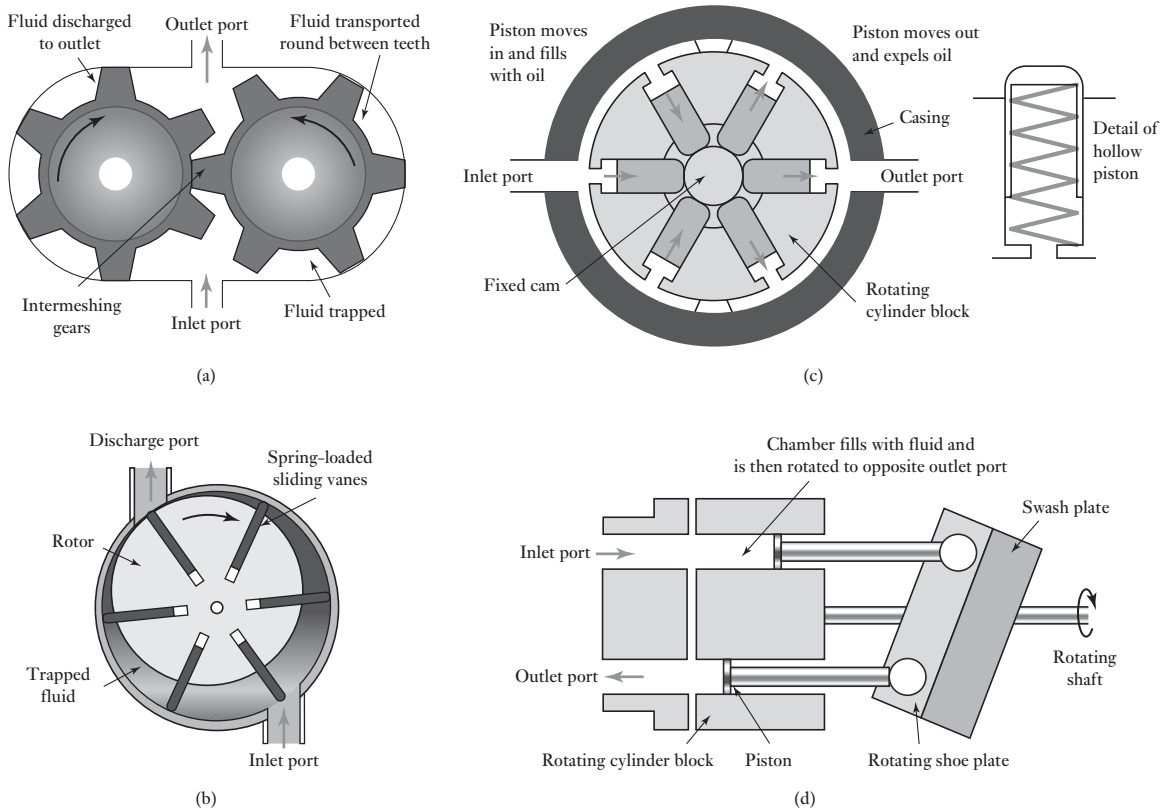
With a hydraulic system, pressurised oil is provided by a pump driven by an electric motor. The pump pumps oil from a sump through a non-return

**Figure 7.1** (a) Hydraulic power supply, (b) accumulator.



valve and an accumulator to the system, from which it returns to the sump. Figure 7.1(a) illustrates the arrangement. A pressure-relief valve is included, this being to release the pressure if it rises above a safe level, the non-return valve is to prevent the oil being back driven to the pump and the accumulator is to smooth out any short-term fluctuations in the output oil pressure. Essentially the accumulator is just a container in which the oil is held under pressure against an external force, Figure 7.1(b) showing the most commonly used form which is gas pressurised and involves gas within a bladder in the chamber containing the hydraulic fluid; an older type involved a spring-loaded piston. If the oil pressure rises then the bladder contracts, increases the volume the oil can occupy and so reduces the pressure. If the oil pressure falls, the bladder expands to reduce the volume occupied by the oil and so increases its pressure.

Commonly used hydraulic pumps are the gear pump, the vane pump and the piston pump. The **gear pump** consists of two close-meshing gear wheels which rotate in opposite directions (Figure 7.2(a)). Fluid is forced through the pump as it becomes trapped between the rotating gear teeth and the housing and so is transferred from the inlet port to be discharged at the outlet port. Such pumps are widely used, being low cost and robust. They generally operate at pressures below about 15 MPa and at 2400 rotations per minute. The maximum flow capacity is about  $0.5 \text{ m}^3/\text{min}$ . However, leakage occurs between the teeth and the casing and between the interlocking teeth, and this limits the efficiency. The **vane pump** has spring-loaded sliding vanes slotted in a driven rotor (Figure 7.2(b)). As the rotor rotates, the vanes follow the contours of the casing. This results in fluid becoming trapped between successive vanes and the casing and transported round from the inlet port to outlet port. The leakage is less than with the gear pump. **Piston pumps** used in hydraulics can take a number of forms. With the **radial piston pump** (Figure 7.2(c)), a cylinder block rotates round the stationary cam and this causes hollow pistons, with spring return, to move in and out. The result is that fluid is drawn in from the inlet port and transported round for ejection from the discharge port. The **axial piston pump** (Figure 7.2(d)) has pistons which move axially rather than radially. The pistons are arranged



**Figure 7.2** (a) Gear pump, (b) vane pump, (c) radial piston pump, (d) axial piston pump with swash plate.

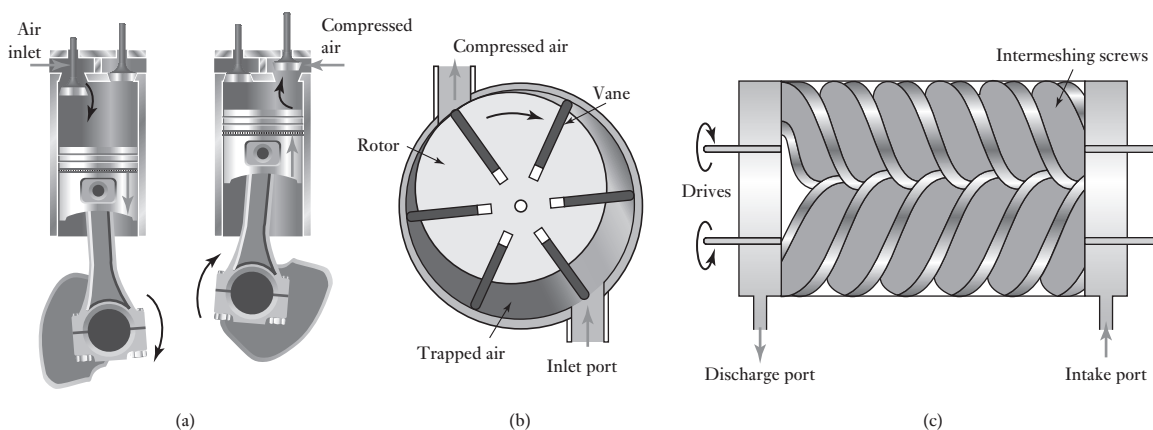
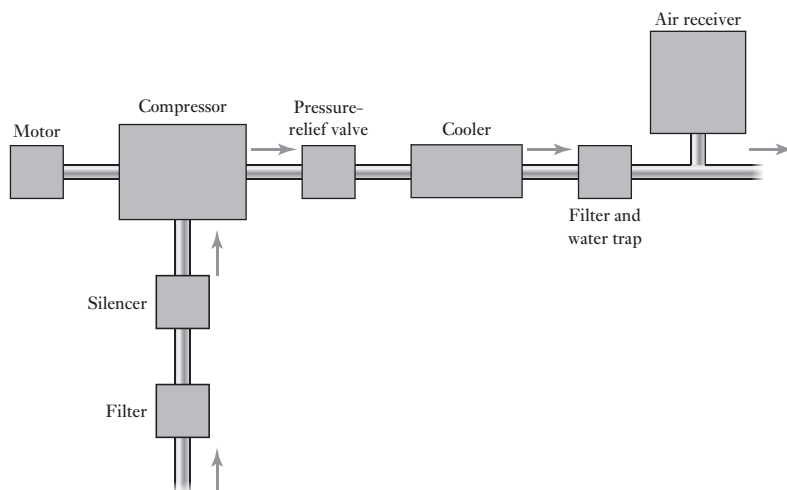
axially in a rotating cylinder block and made to move by contact with the swash plate. This plate is at an angle to the drive shaft and thus as the shaft rotates they move the pistons so that air is sucked in when a piston is opposite the inlet port and expelled when it is opposite the discharge port. Piston pumps have a high efficiency and can be used at higher hydraulic pressures than gear or vane pumps.

### 7.2.2 Pneumatic systems

With a **pneumatic** power supply (Figure 7.3) an electric motor drives an air compressor. The air inlet to the compressor is likely to be filtered and via a silencer to reduce the noise level. A pressure-relief valve provides protection against the pressure in the system rising above a safe level. Since the air compressor increases the temperature of the air, there is likely to be a cooling system and to remove contamination and water from the air a filter with a water trap. An air receiver increases the volume of air in the system and smoothes out any short-term pressure fluctuations.

Commonly used air compressors are ones in which successive volumes of air are isolated and then compressed. Figure 7.4(a) shows the basic form of

**Figure 7.3** Pneumatic power supply.



**Figure 7.4** (a) Single-acting, single stage, vertical, reciprocating compressor, (b) rotary vane compressor, (c) screw compressor.

a single-acting, single-stage, vertical, reciprocating compressor. On the air intake stroke, the descending piston causes air to be sucked into the chamber through the spring-loaded inlet valve and when the piston starts to rise again, the trapped air forces the inlet valve to close and so becomes compressed. When the air pressure has risen sufficiently, the spring-loaded outlet valve opens and the trapped air flows into the compressed-air system. After the piston has reached the top dead centre it then begins to descend and the cycle repeats itself. Such a compressor is termed **single-acting** because one pulse of air is produced per piston stroke; **double-acting** compressors are designed to produce pulses of air on both the up and down strokes of the piston. It is also termed **single-stage** because the compressor goes directly from atmospheric pressure to the required pressure in a single operation. For the production of compressed air at more than a few bars, two or more stages are generally used. Normally two stages are used for pressures up to about 10 to 15 bar and more stages for higher pressures. Thus with a two-stage

compressor we might have the first stage taking air at atmospheric pressure and compressing it to, say, 2 bar and then the second stage compressing this air to, say, 7 bar. Reciprocating piston compressors can be used as a single-stage compressor to produce air pressures up to about 12 bar and as a multistage compressor up to about 140 bar. Typically, air flow deliveries tend to range from about 0.02 m<sup>3</sup>/min free air delivery to about 600 m<sup>3</sup>/min free air delivery; free air is the term used for air at normal atmospheric pressure. Another form of compressor is the **rotary vane compressor**. This has a rotor mounted eccentrically in a cylindrical chamber (Figure 7.4(b)). The rotor has blades, the vanes, which are free to slide in radial slots with rotation causing the vanes to be driven outwards against the walls of the cylinder. As the rotor rotates, air is trapped in pockets formed by the vanes and as the rotor rotates so the pockets become smaller and the air is compressed. Compressed packets of air are thus discharged from the discharge port. Single-stage, rotary vane compressors typically can be used for pressures up to about 800 kPa with flow rates of the order of 0.3 m<sup>3</sup>/min to 30 m<sup>3</sup>/min free air delivery. Another form of compressor is the **rotary screw compressor** (Figure 7.4(c)). This has two intermeshing rotary screws which rotate in opposite directions. As the screws rotate, air is drawn into the casing through the inlet port and into the space between the screws. Then this trapped air is moved along the length of the screws and compressed as the space becomes progressively smaller, emerging from the discharge port. Typically, single-stage, rotary screw compressors can be used for pressures up to about 1000 kPa with flow rates of between 1.4 m<sup>3</sup>/min and 60 m<sup>3</sup>/min free air delivery.

### 7.2.3 Valves

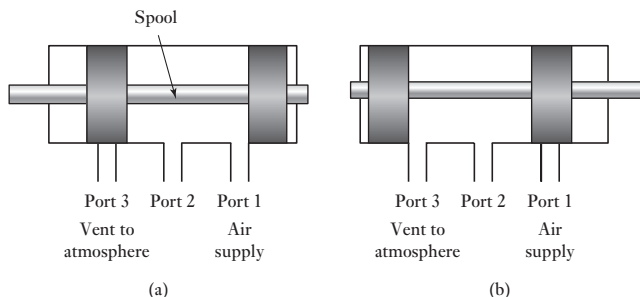
Valves are used with hydraulic and pneumatic systems to direct and regulate the fluid flow. There are basically just two forms of valve, the **finite position** and the **infinite position** valves. The finite position valves are ones where the action is just to allow or block fluid flow and so can be used to switch actuators on or off. They can be used for directional control to switch the flow from one path to another and so from one actuator to another. The infinite position valves are able to control flow anywhere between fully on and fully off and so are used to control varying actuator forces or the rate of fluid flow for a process control situation.

## 7.3

### Directional control valves

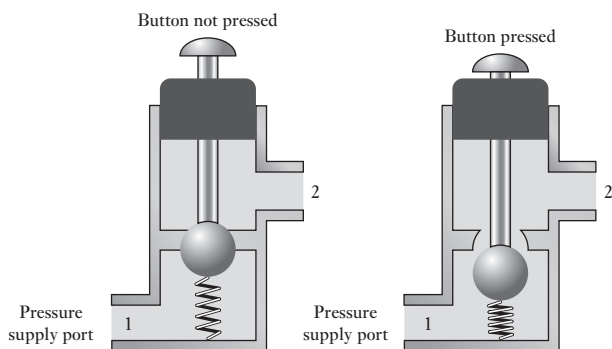
Pneumatic and hydraulic systems use directional control valves to direct the flow of fluid through a system. They are not intended to vary the rate of flow of fluid but are either completely open or completely closed, i.e. on/off devices. Such on/off valves are widely used to develop sequenced control systems (see Section 7.5.1). They might be activated to switch the fluid flow direction by means of mechanical, electrical or fluid pressure signals.

A common type of directional control valve is the **spool valve**. A spool moves horizontally within the valve body to control the flow. Figure 7.5

**Figure 7.5** Spool valve.

shows a particular form. In (a) the air supply is connected to port 1 and port 3 is closed. Thus the device connected to port 2 can be pressurised. When the spool is moved to the left (Figure 7.5(b)) the air supply is cut off and port 2 is connected to port 3. Port 3 is a vent to the atmosphere and so the air pressure in the system attached to port 2 is vented. Thus the movement of the spool has allowed the air firstly to flow into the system and then be reversed and flow out of the system. **Rotary spool valves** have a rotating spool which, when it rotates, opens and closes ports in a similar way.

Another common form of directional control valve is the **poppet valve**. Figure 7.6 shows one form. This valve is normally in the closed condition, there being no connection between port 1 to which the pressure supply is connected and port 2 to which the system is connected. In poppet valves, balls, discs or cones are used in conjunction with valve seats to control the flow. In the figure a ball is shown. When the push-button is depressed, the ball is pushed out of its seat and flow occurs as a result of port 1 being connected to port 2. When the button is released, the spring forces the ball back up against its seat and so closes off the flow.

**Figure 7.6** Poppet valve.

### 7.3.1 Valve symbols

The symbol used for a control valve consists of a square for each of its switching positions. Thus for the poppet valve shown in Figure 7.6, there

are two positions: one with the button not pressed and one with it pressed. Thus a two-position valve will have two squares, a three-position valve three squares. Arrow-headed lines (Figure 7.7(a)) are used to indicate the directions of flow in each of the positions, with blocked-off lines indicating closed flow lines (Figure 7.7(b)). The initial position of the valve has the connections (Figure 7.7(c)) to the ports shown; in Figure 7.7(c) the valve has four ports. Ports are labelled by a number or a letter according to their function. The ports are labelled 1 (or P) for pressure supply, 3 (or T) for hydraulic return port, 3 or 5 (or R or S) for pneumatic exhaust ports, and 2 or 5 (or B or A) for output ports.

**Figure 7.7** (a) Flow path, (b) flow shut-off, (c) initial connections.

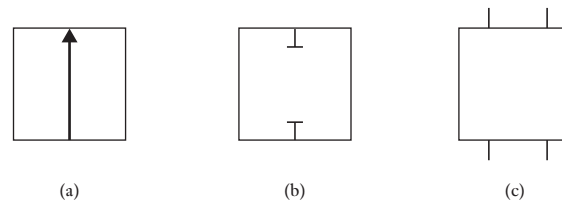
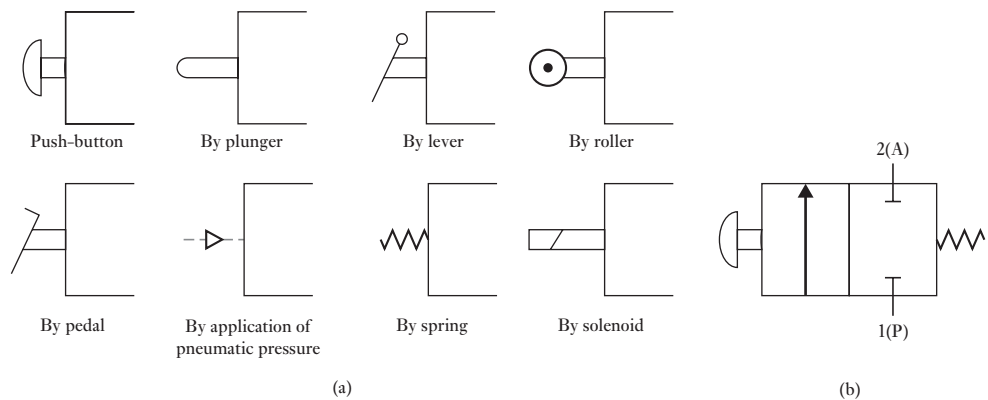


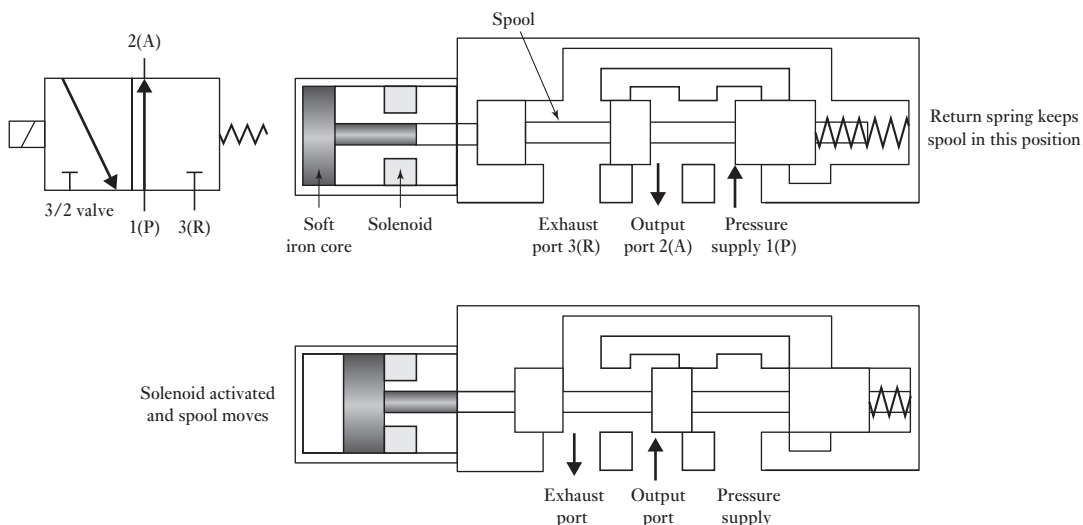
Figure 7.8(a) shows examples of some of the symbols which are used to indicate the various ways the valves can be actuated. More than one of these symbols might be used with the valve symbol. As an illustration, Figure 7.8(b) shows the symbol for the two-port, two-position poppet valve of Figure 7.6. Note that a two-port, two-position valve would be described as a 2/2 valve, the first number indicating the number of ports and the second number the number of positions. The valve actuation is by a push-button and a spring.



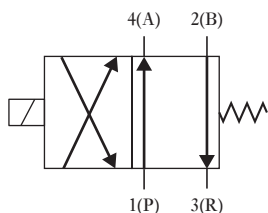
**Figure 7.8** Valve actuation symbols.

As a further illustration, Figure 7.9 shows a solenoid-operated spool valve and its symbol. The valve is actuated by a current passing through a solenoid and returned to its original position by a spring.

Figure 7.10 shows the symbol for a 4/2 valve. The connections are shown for the initial state, i.e. 1(P) is connected to 2(A) and 3(R) closed. When the



**Figure 7.9** Single-solenoid valve.

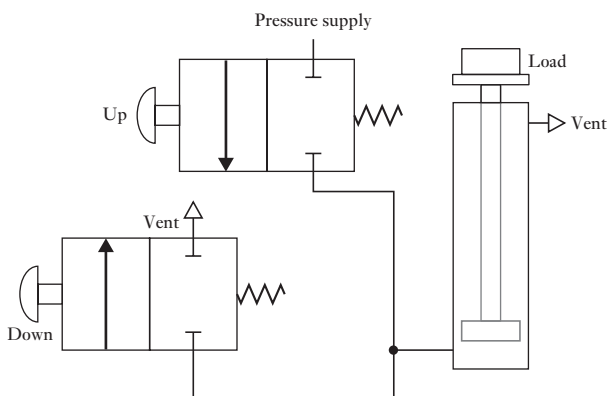


**Figure 7.10** A 4/2 valve.

solenoid is activated it gives the state indicated by the symbols used in the square to which it is attached, i.e. we now have 1(P) closed and 2(A) connected to 3(R). When the current through the solenoid ceases, the spring pushes the valve back to its initial position. The spring movement gives the state indicated by the symbols used in the square to which it is attached.

Figure 7.11 shows a simple example of an application of valves in a pneumatic lift system. Two push-button 2/2 valves are used. When the button on the up valve is pressed, the load is lifted. When the button on the down valve is pressed, the load is lowered. Note that with pneumatic systems an open arrow is used to indicate a vent to the atmosphere.

**Figure 7.11** Lift system.

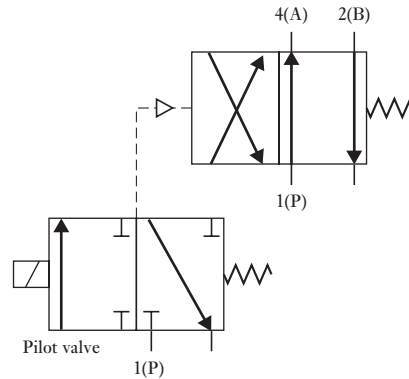


### 7.3.2 Pilot-operated valves

The force required to move the ball or shuttle in a valve can often be too large for manual or solenoid operation. To overcome this problem a **pilot-operated**



**Figure 7.12** Pilot-operated system.

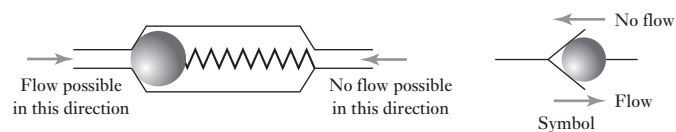


**system** is used where one valve is used to control a second valve. Figure 7.12 illustrates this. The pilot valve is small capacity and can be operated manually or by a solenoid. It is used to allow the main valve to be operated by the system pressure. The pilot pressure line is indicated by dashes. The pilot and main valves can be operated by two separate valves but they are often combined in a single housing.

### 7.3.3 Directional valves

Figure 7.13 shows a simple **directional valve** and its symbol. Free flow can only occur in one direction through the valve: that which results in the ball being pressed against the spring. Flow in the other direction is blocked by the spring forcing the ball against its seat.

**Figure 7.13** Directional valve.



## 7.4 Pressure control valves

There are three main types of pressure control valves.

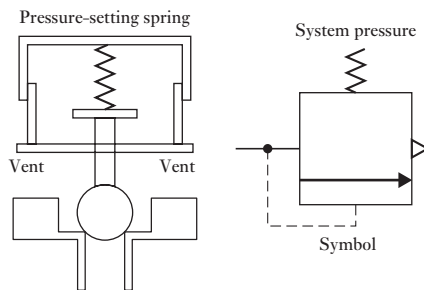
### 1 *Pressure-regulating valves*

These are used to control the operating pressure in a circuit and maintain it at a constant value.

### 2 *Pressure-limiting valves*

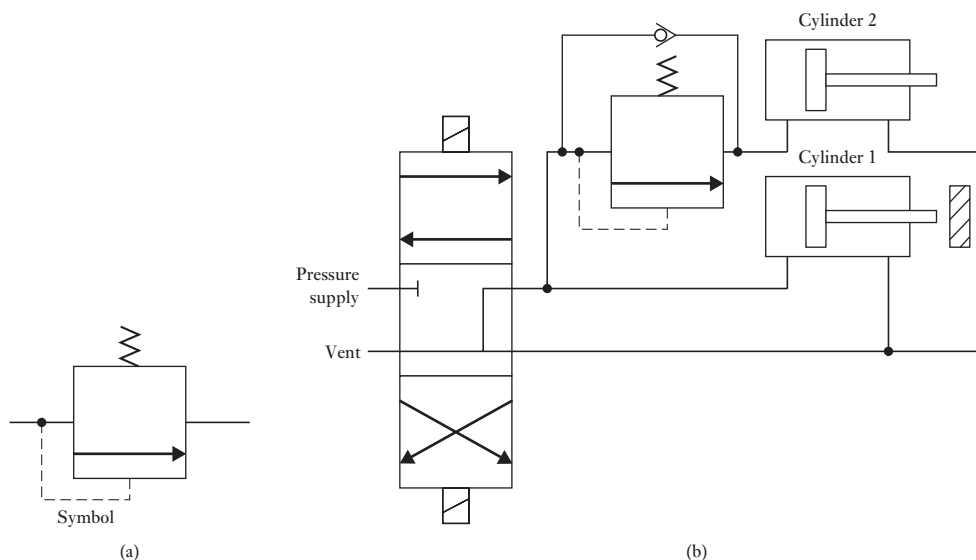
These are used as safety devices to limit the pressure in a circuit to below some safe value. The valve opens and vents to the atmosphere, or back to the sump, if the pressure rises above the set safe value. Figure 7.14 shows a **pressure-limiting/relief valve** which has one orifice which is normally closed. When the inlet pressure overcomes the force exerted by the spring, the valve opens and vents to the atmosphere, or back to the sump.

**Figure 7.14** Pressure-limiting valve.

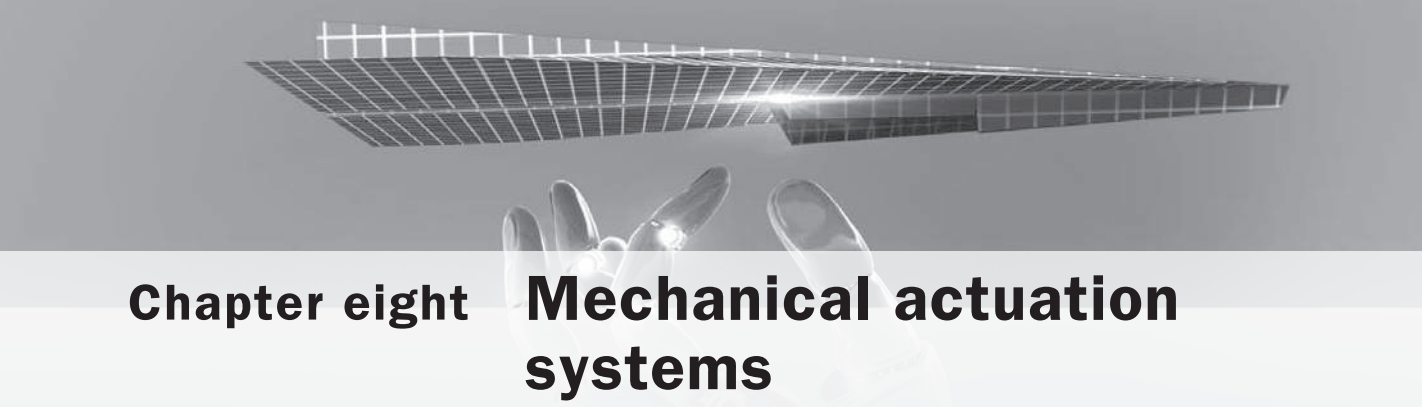


### 3 Pressure sequence valves

These valves are used to sense the pressure of an external line and give a signal when it reaches some preset value. With the pressure-limiting valve of Figure 7.15, the limiting pressure is set by the pressure at the inlet to the valve. We can adapt such a valve to give a sequence valve. This can be used to allow flow to occur to some part of the system when the pressure has risen to the required level. For example, in an automatic machine we might require some operation to start when the clamping pressure applied to a workpiece is at some particular value. Figure 7.15(a) shows the symbol for a sequence valve, the valve switching on when the inlet pressure reaches a particular value and allowing the pressure to be applied to the system that follows. Figure 7.15(b) shows a system where such a sequential valve is used. When the 4/3 valve first operates, the pressure is applied to cylinder 1 and its ram moves to the right. While this is happening the pressure is too low to operate the sequence valve and so no pressure is applied to cylinder 2. When the ram of cylinder 1 reaches the end stop, then the pressure in the system rises and, at an appropriate level, triggers the sequence valve to open and so apply pressure to cylinder 2 to start its ram in motion.



**Figure 7.15** (a) Pressure sequence valve symbol, (b) a sequential system.



# Chapter eight Mechanical actuation systems

## Objectives

The objectives of this chapter are that, after studying it, the reader should be able to:

- Determine possible mechanical actuation systems for motion transmission involving linear-to-rotary, rotary-to-rotary, rotary-to-linear and cyclic motion transmission.
- Evaluate the capabilities of linkages, cams, gears, ratchet-and-pawl, belt and chain drives and bearings for actuation systems.

### 8.1

#### Mechanical systems

This chapter is a consideration of **mechanisms**: mechanisms are devices which can be considered to be motion converters in that they transform motion from one form to some other required form. They might, for example, transform linear motion into rotational motion, or motion in one direction into a motion in a direction at right angles, or perhaps a linear reciprocating motion into rotary motion, as in the internal combustion engine where the reciprocating motion of the pistons is converted into rotation of the crank and hence the drive shaft.

Mechanical elements can include the use of linkages, cams, gears, rack-and-pinion, chains, belt drives, etc. For example, the rack-and-pinion can be used to convert rotational motion to linear motion. Parallel shaft gears might be used to reduce a shaft speed. Bevel gears might be used for the transmission of rotary motion through  $90^\circ$ . A toothed belt or chain drive might be used to transform rotary motion about one axis to motion about another. Cams and linkages can be used to obtain motions which are prescribed to vary in a particular manner. This chapter is a consideration of the basic characteristics of a range of such mechanisms.

Many of the actions which previously were obtained by the use of mechanisms are, however, often nowadays being obtained, as a result of a mechatronics approach, by the use of microprocessor systems. For example, cams on a rotating shaft were previously used for domestic washing machines in order to give a timed sequence of actions such as opening a valve to let water into the drum, switching the water off, switching a heater on, etc. Modern washing machines use a microprocessor-based system with the microprocessor programmed to switch on outputs in the required sequence. Another example is the hairspring balance wheel and gears and pointer of a watch which have now largely been replaced by an integrated circuit with perhaps a liquid crystal display. The mechatronics approach has resulted in a simplification, and often a reduction in cost.

Mechanisms still, however, have a role in mechatronics systems. For example, the mechatronics system in use in an automatic camera for adjusting the aperture for correct exposures involves a mechanism for adjusting the size of the diaphragm.

While electronics might now be used often for many functions that previously were fulfilled by mechanisms, mechanisms might still be used to provide such functions as:

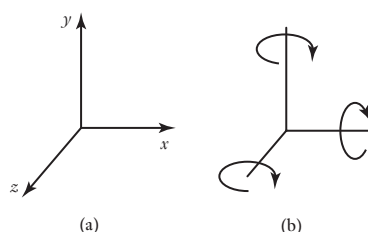
- 1 force amplification, e.g. that given by levers;
- 2 change of speed, e.g. that given by gears;
- 3 transfer of rotation about one axis to rotation about another, e.g. a timing belt;
- 4 particular types of motion, e.g. that given by a quick-return mechanism.

The term **kinematics** is used for the study of motion without regard to forces. When we consider just the motions without any consideration of the forces or energy involved then we are carrying out a kinematic analysis of the mechanism. This chapter is an introduction to such a consideration.

## 8.2 Types of motion

The motion of any rigid body can be considered to be a combination of translational and rotational motions. By considering the three dimensions of space, a **translation motion** can be considered to be a movement which can be resolved into components along one or more of the three axes (Figure 8.1(a)). A **rotational motion** can be considered as a rotation which has components rotating about one or more of the axes (Figure 8.1(b)).

**Figure 8.1** Types of motion:  
(a) translational, (b) rotational.

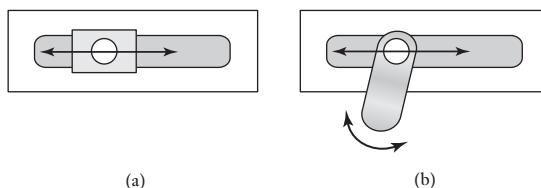


A complex motion may be a combination of translational and rotational motions. For example, think of the motion required for you to pick up a pencil from a table. This might involve your hand moving at a particular angle towards the table, rotation of the hand, and then all the movement associated with opening your fingers and moving them to the required positions to grasp the pencil. This is a sequence of quite complex motions. However, we can break down all these motions into combinations of translational and rotational motions. Such an analysis is particularly relevant if we are not moving a human hand to pick up the pencil but instructing a robot to carry out the task. Then it really is necessary to break down the motion into combinations of translational and rotational motions so that we can design mechanisms to carry out each of these components of the motion. For example, among the sequence of control signals sent to a mechanism might be such groupings of signals as those to instruct joint 1 to rotate by  $20^\circ$  and link 2 to be extended by 4 mm for translational motion.

### 8.2.1 Freedom and constraints

An important aspect in the design of mechanical elements is the orientation and arrangement of the elements and parts. A body that is free in space can move in three, independent, mutually perpendicular directions and rotate in three ways about those directions (Figure 8.1). It is said to have six degrees of freedom. The number of **degrees of freedom** is the number of components of motion that are required in order to generate the motion. If a joint is constrained to move along a line then its translational degrees of freedom are reduced to one. Figure 8.2(a) shows a joint with just this one translational degree of freedom. If a joint is constrained to move on a plane then it has two translational degrees of freedom. Figure 8.2(b) shows a joint which has one translational degree of freedom and one rotational degree of freedom.

**Figure 8.2** Joints with: (a) one, (b) two degrees of freedom.



The problem in design is often to reduce the number of degrees of freedom and this then requires an appropriate number and orientation of constraints. Without any constraints a body would have six degrees of freedom. A constraint is needed for each degree of freedom that is to be prevented from occurring. Provided we have no redundant constraints then the number of degrees of freedom would be 6 minus the number of constraints. However, redundant constraints often occur and so for constraints on a single rigid body we have the basic rule

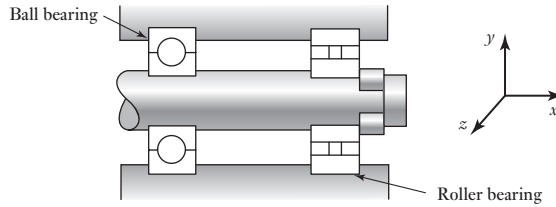
$$6 - \text{number of constraints} = \text{number of degrees of freedom} \\ - \text{number of redundancies}$$

Thus if a body is required to be fixed, i.e. have zero degrees of freedom, then if no redundant constraints are introduced the number of constraints required is 6.

A concept that is used in design is that of the **principle of least constraint**. This states that in fixing a body or guiding it to a particular type of motion, the minimum number of constraints should be used, i.e. there should be no redundancies. This is often referred to as **kinematic design**.

For example, to have a shaft which only rotates about one axis with no translational motions, we have to reduce the number of degrees of freedom to 1. Thus the minimum number of constraints to do this is 5. Any more constraints than this will give redundancies. The mounting that might be used to mount the shaft has a ball bearing at one end and a roller bearing at the other (Figure 8.3). The pair of bearings together prevent translational motion at right angles to the shaft, the  $y$ -axis, and rotations about the  $z$ -axis and the  $y$ -axis. The ball bearing prevents translational motion along the  $x$ -axis and along the  $z$ -axis. Thus there is a total of five constraints. This leaves just one degree of freedom, the required rotation about the  $x$ -axis. If

**Figure 8.3** Shaft with no redundancies.



there had been a roller bearing at each end of the shaft then both the bearings could have prevented translational motion along the  $x$ -axis and the  $z$ -axis and thus there would have been redundancy. Such redundancy might cause damage. If ball bearings are used at both ends of the shaft, then in order to prevent redundancy one of the bearings would have its outer race not fixed in its housing so that it could slide to some extent in an axial direction.

### 8.2.2 Loading

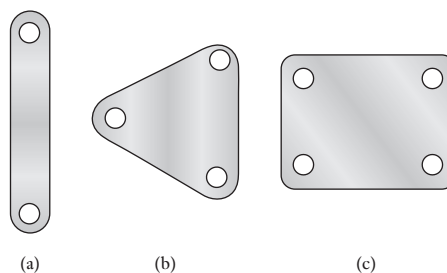
Mechanisms are structures and as such transmit and support loads. Analysis is thus necessary to determine the loads to be carried by individual elements. Then consideration can be given to the dimensions of the element so that it might, for example, have sufficient strength and perhaps stiffness under such loading.

## 8.3

### Kinematic chains

When we consider the movements of a mechanism without any reference to the forces involved, we can treat the mechanism as being composed of a series of individual links. Each part of a mechanism which has motion relative to some other part is termed a **link**. A link need not necessarily be a rigid body but it must be a resistant body which is capable of transmitting the required force with negligible deformation. For this reason it is usually taken as being represented by a rigid body which has two or more points of attachment to other links, these being termed **nodes**. Each link is capable of moving relative to its neighbouring links. Figure 8.4 shows examples of links with two, three and four nodes. A **joint** is a connection between two or more links at their nodes and which allows some motion between the connected links. Levers, cranks, connecting rods and pistons, sliders, pulleys, belts and shafts are all examples of links.

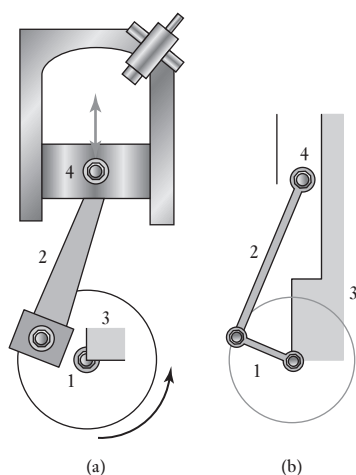
**Figure 8.4** Links: (a) with two nodes, (b) with three nodes, (c) with four nodes.



A sequence of joints and links is known as a **kinematic chain**. For a kinematic chain to transmit motion, one link must be fixed. Movement of one link will then produce predictable relative movements of the others. It is possible to obtain from one kinematic chain a number of different mechanisms by having a different link as the fixed one.

As an illustration of a kinematic chain, consider a motor car engine where the reciprocating motion of a piston is transformed into rotational motion of a crankshaft on bearings mounted in a fixed frame (Figure 8.5(a)). We can represent this as being four connected links (Figure 8.5(b)). Link 1 is the crankshaft, link 2 the connecting rod, link 3 the fixed frame and link 4 the slider, i.e. piston, which moves relative to the fixed frame (see Section 8.3.2 for further discussion).

**Figure 8.5** Simple engine mechanism.

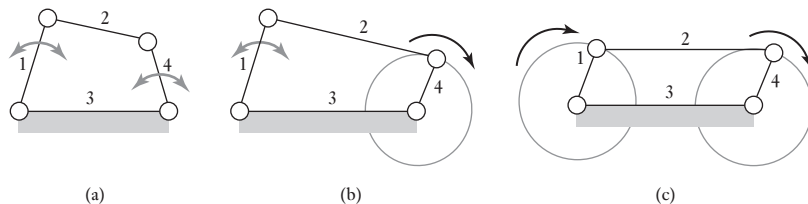


The designs of many mechanisms are based on two basic forms of kinematic chains, the four-bar chain and the slider–crank chain. The following illustrates some of the forms such chains can take.

### 8.3.1 The four-bar chain

The **four-bar chain** consists of four links connected to give four joints about which turning can occur. Figure 8.6 shows a number of forms of the four-bar chain produced by altering the relative lengths of the links. If the

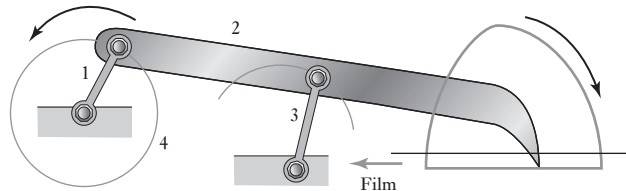
**Figure 8.6** Examples of four-bar chains.



sum of the length of the shortest link plus the length of the longest link is less than or equal to the sum of the lengths of the other two links then at least one link will be capable of making a full revolution with respect to the fixed link. If this condition is not met then no link is capable of a complete revolution. This is known as the Grashof condition. In Figure 8.6(a), link 3 is fixed and the relative lengths of the links are such that links 1 and 4 can oscillate but not rotate. The result is a **double-lever mechanism**. By shortening link 4 relative to link 1, then link 4 can rotate (Figure 8.6(b)) with link 1 oscillating and the result is termed a **lever-crank mechanism**. With links 1 and 4 the same length and both able to rotate (Figure 8.6(c)), then the result is a **double-crank mechanism**. By altering which link is fixed, other forms of mechanism can be produced.

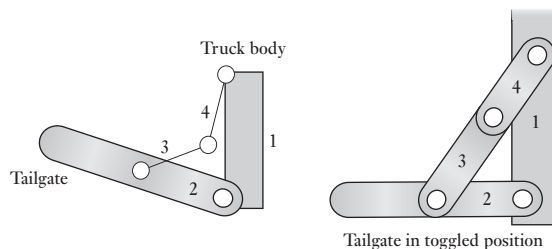
Figure 8.7 illustrates how such a mechanism can be used to advance the film in a cine camera. As link 1 rotates so the end of link 2 locks into a sprocket of the film, pulls it forward before releasing and moving up and back to lock into the next sprocket.

**Figure 8.7** Cine film advance mechanism.



Some linkages may have **toggle positions**. These are positions where the linkage will not react to any input from one of its links. Figure 8.8 illustrates such a toggle, being the linkage used to control the movement of the tailgate of a truck so that when link 2 reaches the horizontal position no further load on link 2 will cause any further movement. There is another toggle position for the linkage and that is when links 3 and 4 are both vertical and the tailgate is vertical.

**Figure 8.8** Toggle linkage.

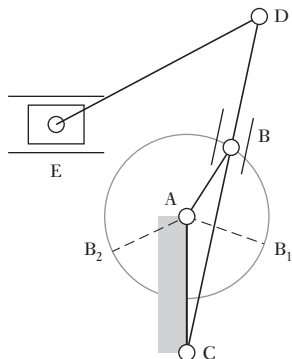
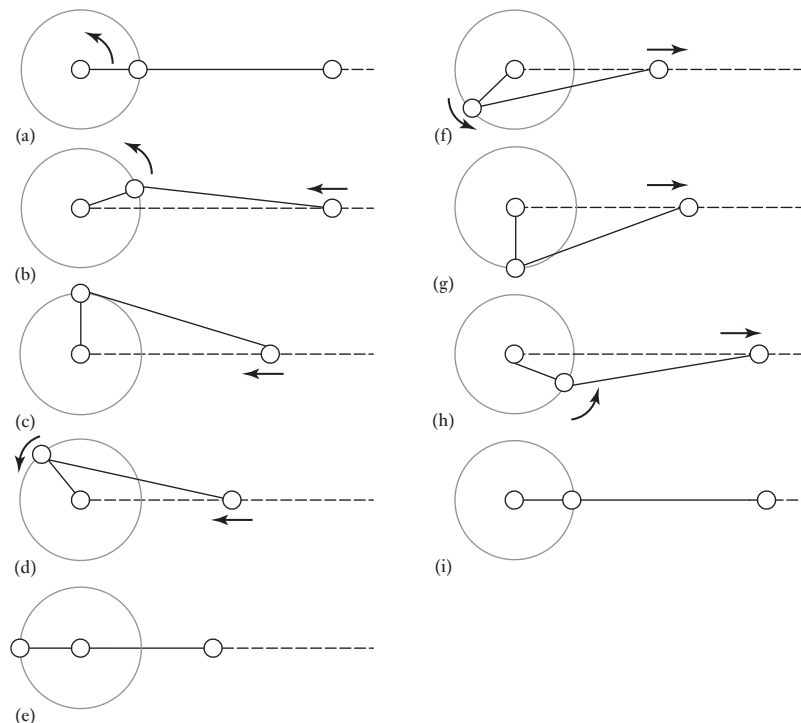


### 8.3.2 The slider-crank mechanism

This form of mechanism consists of a crank, a connecting rod and a slider and is the type of mechanism described in Figure 8.5 which showed the simple engine mechanism. With that configuration, link 3 is fixed, i.e. there is no relative movement between the centre of rotation of the crank and the housing in which the piston slides. Link 1 is the crank that rotates, link 2 the connecting rod and link 4 the slider which moves relative to the fixed link. When the piston moves backwards and forwards, i.e. link 4 moves



**Figure 8.9** The position sequence for the links in a slider-crank mechanism.



**Figure 8.10** Quick-return mechanism.

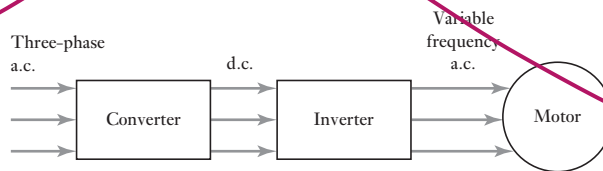
backwards and forwards, then the crank, link 1, is forced to rotate. Hence the mechanism transforms an input of backwards and forwards motion into rotational motion. Figure 8.9 shows a number of stages in this motion. A useful way of seeing how any mechanism might behave is to construct, to scale, a cardboard model and move the links. Changing the length of a link then enables the changes in behaviour of the mechanism to be determined.

Figure 8.10 shows another form of this type of mechanism, a **quick-return mechanism**. It consists of a rotating crank, link AB, which rotates round a fixed centre, an oscillating lever CD, which is caused to oscillate about C by the sliding of the block at B along CD as AB rotates, and a link DE which causes E to move backwards and forwards. E might be the ram of a machine and have a cutting tool attached to it. The ram will be at the extremes of its movement when the positions of the crank are  $AB_1$  and  $AB_2$ . Thus, as the crank moves anti-clockwise from  $B_1$  to  $B_2$ , the ram makes a complete stroke, the cutting stroke. When the crank continues its movement from  $B_2$  anti-clockwise to  $B_1$  the ram again makes a complete stroke in the opposite direction, the return stroke. With the crank rotating at constant speed, because the angle of crank rotation required for the cutting stroke is greater than the angle for the return stroke, the cutting stroke takes more time than the return stroke hence the term quick-return for the mechanism. Similar diagrams, and a cardboard model, can be constructed in the same way as that shown in Figure 8.9.

A **cam** is a body which rotates or oscillates and in doing so imparts a reciprocating or oscillatory motion to a second body, called the **follower**, with which it is in contact (Figure 8.11). As the cam rotates so the follower is made to rise, dwell and fall; the lengths of times spent at each of these

dropping as a result of technological developments and the reduction in price of solid-state devices. Speed control of a.c. motors is based around the provision of a variable frequency supply, since the speed of such motors is determined by the frequency of the supply. The torque developed by an a.c. motor is constant when the ratio of the applied stator voltage to frequency is constant. Thus to maintain a constant torque at the different speeds when the frequency is varied, the voltage applied to the stator has also to be varied. With one method, the alternating current is first rectified to direct current by a **converter** and then **inverted** back to alternating current again but at a frequency that can be selected (Figure 9.23). Another method that is often used for operating slow-speed motors is the **cycloconverter**. This converts alternating current at one frequency directly to alternating current at another frequency without the intermediate d.c. conversion.

**Figure 9.23** Variable speed a.c. motor.



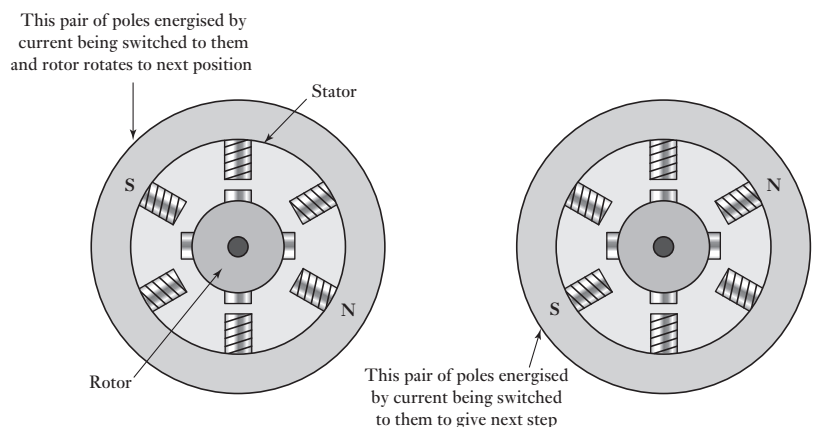
## 9.7 Stepper motors

The **stepper motor** is a device that produces rotation through equal angles, the so-called **steps**, for each digital pulse supplied to its input. Thus, for example, if with such a motor 1 pulse produces a rotation of  $6^\circ$  then 60 pulses will produce a rotation through  $360^\circ$ . There are a number of forms of stepper motor.

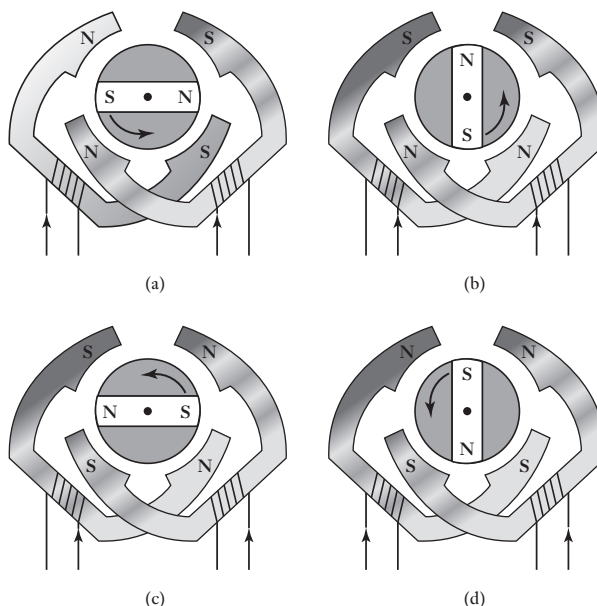
### 1 Variable reluctance stepper

Figure 9.24 shows the basic form of the variable reluctance stepper motor. With this form the rotor is made of soft steel and is cylindrical with four poles, i.e. fewer poles than on the stator. When an opposite pair of windings has current switched to them, a magnetic field is produced with lines of force which pass from the stator poles through the nearest set of poles on the rotor. Since lines of force can be considered to be

**Figure 9.24** Variable reluctance stepper motor.



**Figure 9.25** Permanent magnet two-phase stepper motor with  $90^\circ$  steps. (a), (b), (c) and (d) show the positions of the magnet rotor as the coils are energised in different directions.



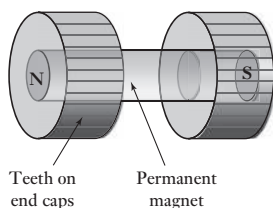
rather like elastic thread and always trying to shorten themselves, the rotor will move until the rotor and stator poles line up. This is termed the position of minimum reluctance. This form of stepper generally gives step angles of  $7.5^\circ$  or  $15^\circ$ .

### 2 Permanent magnet stepper

Figure 9.25 shows the basic form of the permanent magnet motor. The motor shown has a stator with four poles. Each pole is wound with a field winding, the coils on opposite pairs of poles being in series. Current is supplied from a d.c. source to the windings through switches. The rotor is a permanent magnet and thus when a pair of stator poles has a current switched to it, the rotor will move to line up with it. Thus for the currents giving the situation shown in the figure, the rotor moves to the  $45^\circ$  position. If the current is then switched so that the polarities are reversed, the rotor will move a further  $45^\circ$  in order to line up again. Thus by switching the currents through the coils, the rotor rotates in  $45^\circ$  steps. With this type of motor, step angles are commonly  $1.8^\circ$ ,  $7.5^\circ$ ,  $15^\circ$ ,  $30^\circ$ ,  $34^\circ$  or  $90^\circ$ .

### 3 Hybrid stepper

Hybrid stepper motors combine the features of both the variable reluctance and permanent magnet motors, having a permanent magnet encased in iron caps which are cut to have teeth (Figure 9.26). The rotor sets itself in the minimum reluctance position in response to a pair of stator coils being energised. Typical step angles are  $0.9^\circ$  and  $1.8^\circ$ . If a motor has  $n$  phases on the stator and  $m$  teeth on the rotor, the total number of steps per revolution is  $nm$ . Such stepper motors are extensively used in high-accuracy positioning applications, e.g. in computer hard disk drives.



**Figure 9.26** Hybrid motor rotor.

### 9.7.1 Stepper motor specifications

The following are some of the terms commonly used in specifying stepper motors.

1 *Phase*

This term refers to the number of independent windings on the stator, e.g. a four-phase motor. The current required per phase and its resistance and inductance will be specified so that the controller switching output is specified. Two-phase motors, e.g. Figure 9.25, tend to be used in light-duty applications, three-phase motors tend to be variable reluctance steppers, e.g. Figure 9.24, and four-phase motors tend to be used for higher power applications.

2 *Step angle*

This is the angle through which the rotor rotates for one switching change for the stator coils.

3 *Holding torque*

This is the maximum torque that can be applied to a powered motor without moving it from its rest position and causing spindle rotation.

4 *Pull-in torque*

This is the maximum torque against which a motor will start, for a given pulse rate, and reach synchronism without losing a step.

5 *Pull-out torque*

This is the maximum torque that can be applied to a motor, running at a given stepping rate, without losing synchronism.

6 *Pull-in rate*

This is the maximum switching rate at which a loaded motor can start without losing a step.

7 *Pull-out rate*

This is the switching rate at which a loaded motor will remain in synchronism as the switching rate is reduced.

8 *Slew range*

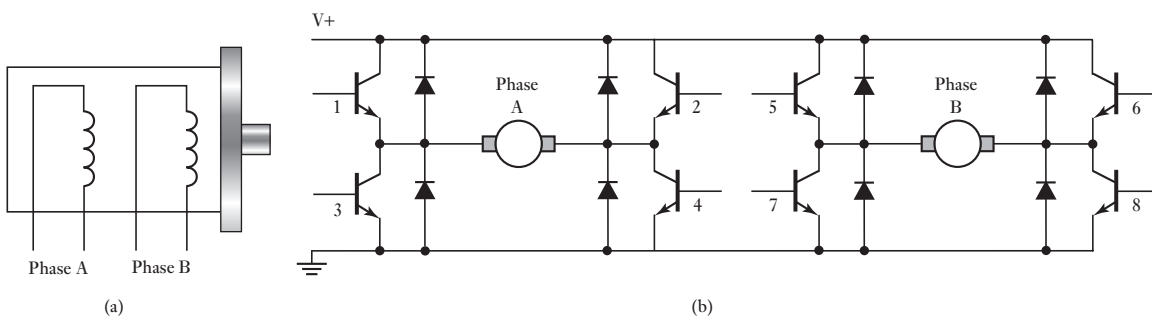
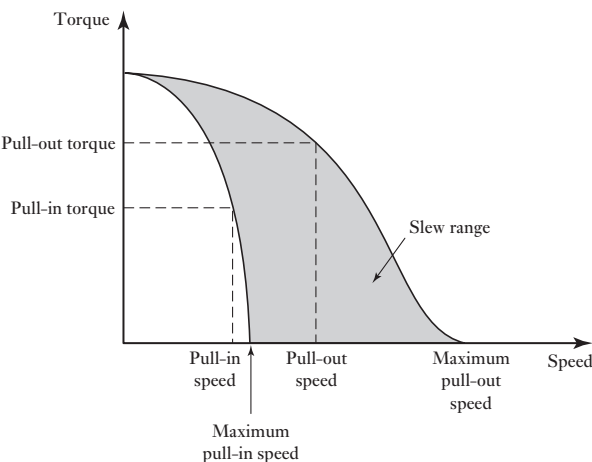
This is the range of switching rates between pull-in and pull-out within which the motor runs in synchronism but cannot start up or reverse.

Figure 9.27 shows the general characteristics of a stepper motor.

### 9.7.2 Stepper motor control

Solid-state electronics is used to switch the d.c. supply between the pairs of stator windings. Two-phase motors, e.g. Figure 9.25, are termed **bipolar motors** when they have four connecting wires for signals to generate the switching sequence (Figure 9.28(a)). Such a motor can be driven by H-circuits (see Figure 9.19 and the associated discussion); Figure 9.28(b) shows the circuit and Table 9.2 shows the switching sequence required for the transistors to carry out the four steps, the sequence then being repeated for

**Figure 9.27** Stepper motor characteristics.



**Figure 9.28** (a) Bipolar motor, (b) H-circuit.

**Table 9.2** Switching sequence for full-stepping bipolar stepper.

| Step | Transistors |         |         |         |
|------|-------------|---------|---------|---------|
|      | 1 and 4     | 2 and 3 | 5 and 8 | 6 and 7 |
| 1    | On          | Off     | On      | Off     |
| 2    | On          | Off     | Off     | On      |
| 3    | Off         | On      | Off     | On      |
| 4    | Off         | On      | On      | Off     |

further steps. The sequence gives a clockwise rotation; for an anti-clockwise rotation the sequence is reversed.

Half-steps, and hence finer resolution, are obtainable if, instead of the full-stepping sequence needed to implement a pole reversal to get from one step to the next, the coils are switched so that the rotor stops at a position halfway to the next full step. Table 9.3 shows the sequence for half-stepping with the bipolar stepper.

Two-phase motors are termed **unipolar** when they have six connecting wires for the generation of the switching sequence (Figure 9.29). Each of the coils has a centre-tap. With the centre-taps of the phase coils connected together, such a form of stepper motor can be switched with just four

**Table 9.3** Half-steps for bipolar stepper.

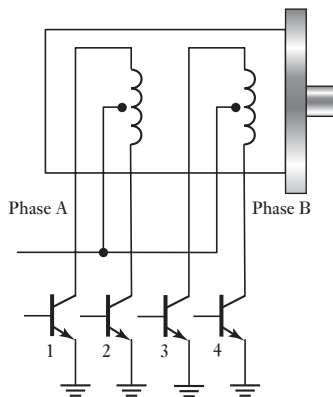
| Step | Transistors |         |         |         |
|------|-------------|---------|---------|---------|
|      | 1 and 4     | 2 and 3 | 5 and 8 | 6 and 7 |
| 1    | On          | Off     | On      | Off     |
| 2    | On          | Off     | Off     | Off     |
| 3    | On          | Off     | Off     | On      |
| 4    | Off         | Off     | Off     | On      |
| 5    | Off         | On      | Off     | On      |
| 6    | Off         | On      | Off     | Off     |
| 7    | Off         | On      | On      | Off     |
| 8    | Off         | Off     | On      | Off     |

**Table 9.4** Switching sequence for full-stepping unipolar stepper.

| Step | Transistors |     |     |     |
|------|-------------|-----|-----|-----|
|      | 1           | 2   | 3   | 4   |
| 1    | On          | Off | On  | Off |
| 2    | On          | Off | Off | On  |
| 3    | Off         | On  | Off | On  |
| 4    | Off         | On  | On  | Off |

**Table 9.5** Half-steps for unipolar stepper.

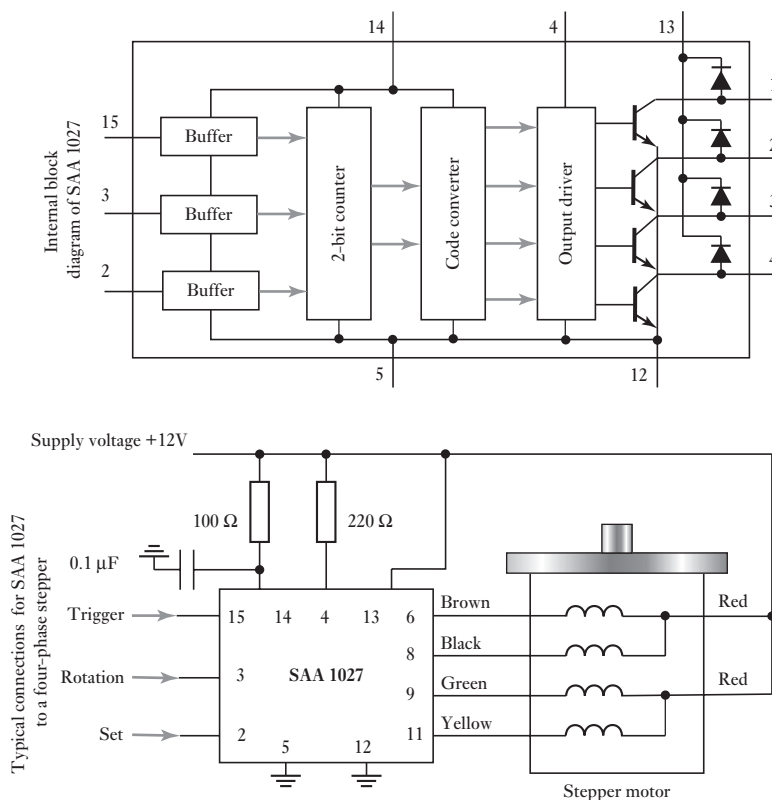
| Step | Transistors |     |     |     |
|------|-------------|-----|-----|-----|
|      | 1           | 2   | 3   | 4   |
| 1    | On          | Off | On  | Off |
| 2    | On          | Off | Off | Off |
| 3    | On          | Off | Off | On  |
| 4    | Off         | Off | Off | On  |
| 5    | Off         | On  | Off | On  |
| 6    | Off         | On  | Off | Off |
| 7    | Off         | On  | On  | Off |
| 8    | Off         | Off | On  | Off |

**Figure 9.29** Unipolar motor.

transistors. Table 9.4 gives the switching sequence for the transistors in order to produce the steps for clockwise rotation, the sequence then being repeated for further steps. For anti-clockwise rotation the sequence is reversed. Table 9.5 shows the sequence when the unipolar is half-stepping.

Integrated circuits are available to provide the drive circuitry. Figure 9.30 shows the connections with the integrated circuit SAA 1027 for a four-phase stepper. The three inputs are controlled by applying high or low signals to them. When the set terminal is held high, the output from the integrated circuit changes state each time the trigger terminal goes from low to high. The sequence repeats itself at four-step intervals but can be reset to the zero condition at any time by applying a low signal to the trigger terminal. When the rotation input is held low there is clockwise rotation, when high it is anti-clockwise.

**Figure 9.30** Integrated circuit SAA 1027 for stepper motor.



Some applications require very small step angles. While the step angle can be made small by increasing the number of rotor teeth and/or the number of phases, generally more than four phases and 50 to 100 teeth are not used. Instead a technique known as **mini-stepping** is used. This involves dividing each step into a number of equal size substeps. This is done by using different currents to the coils so that the rotor moves to intermediate positions between normal step positions. Thus, for example, a step of  $1.8^\circ$  might be subdivided into 10 equal steps.

Stepper motors can be used to give controlled rotational steps but also can give continuous rotation with their rotational speed controlled by controlling the rate at which pulses are applied to it to cause stepping. This gives a very useful controlled variable speed motor which finds many applications.

Because stepper coils have inductance and switched inductive loads can generate large back e.m.f.s when switched, when steppers are connected to microprocessor output ports it is necessary to include protection to avoid damage to the microprocessor. This may take the form of resistors in the lines to limit the current, though these must have values carefully chosen both to provide the protection and not to limit the value of the current needed to switch the transistors. Diodes across the coils prevent current in the reverse direction and so give protection. An alternative is to use optoisolators (see Section 3.3).

### 9.7.3 Selection of a stepper motor

The following should be considered in the selection of a stepper motor.

- 1 The operating torque requirements of the application. The rating torque must be high enough to accommodate the torque and slew range requirement. Also the torque-speed characteristics must be appropriate.
- 2 The step angle must be of high enough resolution to provide the required output motion increments.
- 3 Cost.

This will require looking at the data specifications for stepper motors. The following are some typical values taken from a manufacturer's data sheet for a unipolar stepper motor (Canon 42M048C1U-N):

|                         |                                    |
|-------------------------|------------------------------------|
| d.c. operating voltage  | 5 V                                |
| Resistance per winding  | 9.1 $\Omega$                       |
| Inductance per winding  | 8.1 mH                             |
| Holding torque          | 66.2 mNm/9.4 oz. in                |
| Rotor moment of inertia | $12.5 \times 10^{-4} \text{ gm}^2$ |
| Detent torque           | 12.7 mNm/1.8 oz. in                |
| Step angle              | 7.5°                               |
| Step angle tolerance    | $\pm 0.5^\circ$                    |
| Steps per revolution    | 48                                 |

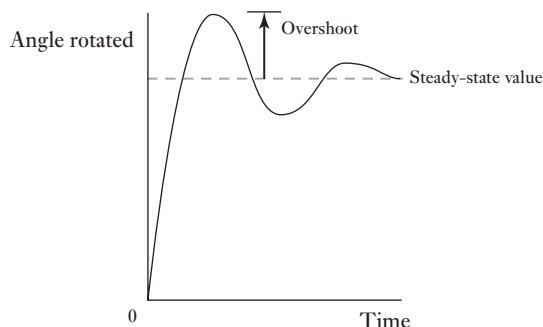
The detent torque is the torque required to rotate the stepper motor when the motor windings are not energised.

Once a motor has been selected, a drive system will need to be found which is compatible with the motor. For example, for use with a unipolar motor the Cybernetics CY512 might be used if a maximum input voltage of 7 V and maximum current per phase of 80 mA is acceptable. The SAA1027 by Signetics is a widely used driver for small unipolar stepper motors with a maximum input voltage of 18 V and a maximum current per phase of 350 mA. For a two-phase bipolar or four-phase unipolar motor, the SCS-Thomson L297/L298 could be considered, it being a two-chip logic driver set. The L297 chip generates the motor phase sequences of four-phase TTL logic signals for two-phase and four-phase unipolar motors and the L298 is a bridge driver designed to accept such signals and drive inductive loads, in this case a stepper. A bipolar motor can be driven with winding currents up to 2 A.

When a pulse is supplied to a stepper motor we have essentially an input to an inductor-resistor circuit and the resulting torque is applied to the load, resulting in angular acceleration. As a consequence, the system will have a natural frequency; it will not go directly to the next step position but will generally have damped oscillations about it before settling down to the steady-state value (Figure 9.31). See Section 24.1.2 for a discussion of this and a derivation of the natural frequency and damping factor.



**Figure 9.31** Oscillations about the steady-state angle.



## 9.8 Motor selection

When selecting a motor for a particular application, factors that need to be taken into account include:

- 1 inertia matching;
- 2 torque requirements;
- 3 power requirements.

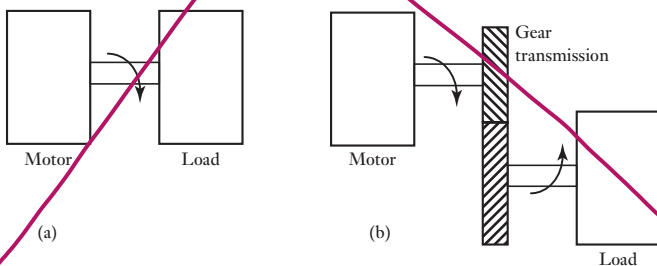
### 9.8.1 Inertia matching

The concept of impedance matching introduced in Section 3.8 for electrical impedances can be extended to mechanical systems, and an analogous situation to that described there for electrical circuits is that of a motor, a torque source, directly rotating a load (Figure 9.32(a)). The torque required to give a load with moment of inertia  $I_L$  and angular acceleration  $\alpha$  is  $I_L\alpha$ . The torque required to accelerate the motor shaft is  $T_M = I_M\alpha_M$  and that required to accelerate the load is  $T_L = I_L\alpha_L$ . The motor shaft will, in the absence of gearing, have the same angular acceleration and the same angular velocity. The power needed to accelerate the system as a whole is  $T_M\omega + T_L\omega$ , where  $\omega$  is the angular velocity. Thus:

$$\text{power} = (I_M + I_L) \alpha \omega$$

This power is produced by the motor torque  $T_M$  and thus the power must equal  $T_M\omega$ . Hence:

$$T = (I_M + I_L) \alpha$$



**Figure 9.32** (a) Motor rotating load, (b) motor with gear transmission rotating load.



# Chapter ten      **Microprocessors and microcontrollers**

## Objectives

The objectives of this chapter are that, after studying it, the reader should be able to:

- Describe the basic structure of a microprocessor system.
- Describe the architecture of common microprocessors and how they can be incorporated in microprocessor systems.
- Describe the basic structure of microcontrollers and how their registers can be set up to carry out tasks.
- Explain how programs can be developed using flow charts and pseudocode.

### 10.1

#### Control

If we take a simple control problem, e.g. the sequencing of the red, amber, green lights at a traffic crossing, then it should be possible to solve it by an electronic control system involving combinational and sequential logic integrated circuits. However, with a more complex situation there might be many more variables to control in a more complex control sequence. The simplest solution now becomes not one of constructing a system based on hard-wired connections of combinational and sequential logic integrated circuits but of using a microprocessor and using software to make the 'interconnections'.

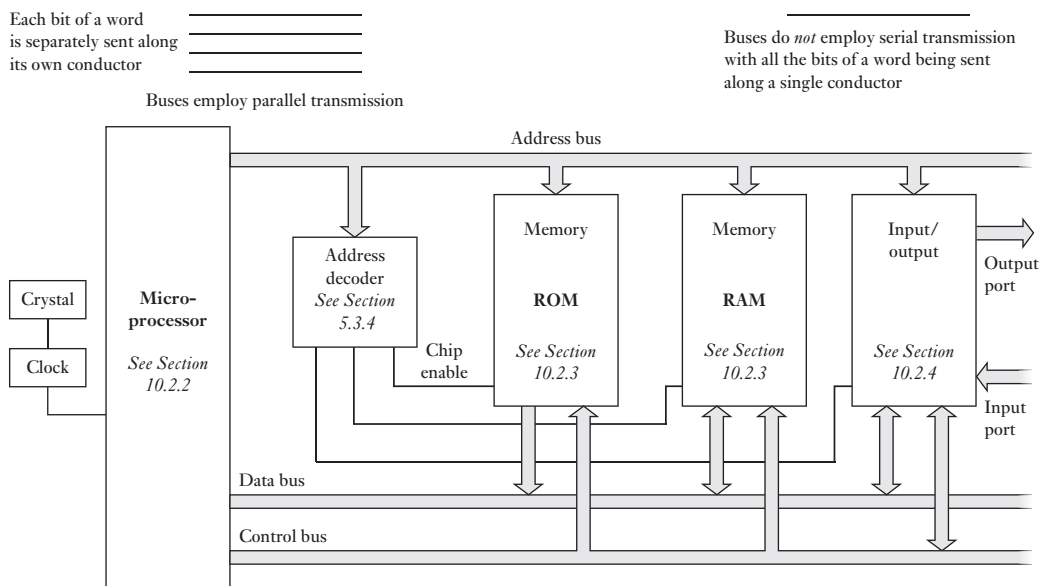
The microprocessor systems that we are concerned with in this book are for use as control systems and are termed **embedded microprocessors**. This is because such a microprocessor is dedicated to controlling a specific function and is self-starting, requiring no human intervention and completely self-contained with its own operating program. For the human, it is not apparent that the system is a microprocessor one. Thus, a modern washing machine contains a microprocessor but all that the operator has to do to operate it is to select the type of wash required by pressing the appropriate button or rotating a switch and then press the button to start.

This chapter is an overview of the structure of microprocessors and microcontrollers with the next two chapters discussing programming and Chapter 13 interfacing.

### 10.2

#### Microprocessor systems

Systems using microprocessors basically have three parts: a **central processing unit** (CPU) to recognise and carry out program instructions (this is the part which uses the microprocessor), **input and output interfaces** to handle communications between the microprocessor and the outside world (the term **port** is used for the interface), and **memory** to



**Figure 10.1** General form of a microprocessor system and its buses. All the components share the same data bus and address bus. This arrangement is known as the von Neumann architecture.

hold the program instructions and data. Figure 10.1 illustrates the general arrangement of a microprocessor system.

Microprocessors which have memory and various input/output arrangements all on the same chip are called **microcontrollers**.

### 10.2.1 Buses

Digital signals move from one section to another along paths called **buses**. A bus, in the physical sense, is just a number of parallel conductors along which electrical signals can be carried and are paths which can be shared by all the chips in the system. This is because if separate connections were used between the chips, there would be a very large number of connecting conductors. Using shared connection buses does mean that when one chip puts data on the bus, the other chips have to wait their turn until the data transfer is complete before one of them can put its data on the bus. Typically a bus has 16 or 32 parallel connections so that each can carry 1 bit of a data word simultaneously. This gives faster transmission than having a serial connection in which an entire word is sent in a sequence of bits along a single conductor.

There are three forms of bus in a microprocessor system.

#### 1 Data bus

The data associated with the processing function of the CPU is carried by the **data bus**. Thus, it is used to transport a word to or from the CPU and the memory or the input/output interfaces. Each wire in the bus carries a binary signal, i.e. a 0 or a 1. Thus with a four-wire bus we might have the

word 1010 being carried, each bit being carried by a separate wire in the bus, as:

| Word                      | Bus wire             |
|---------------------------|----------------------|
| 0 (least significant bit) | First data bus wire  |
| 1                         | Second data bus wire |
| 0                         | Third data bus wire  |
| 1 (most significant bit)  | Fourth data bus wire |

The more wires the data bus has, the longer the word length that can be used. The range of values which a single item of data can have is restricted to that which can be represented by the word length. Thus with a word of length 4 bits the number of values is  $2^4 = 16$ . Thus if the data is to represent, say, a temperature, then the range of possible temperatures must be divided into 16 segments if we are to represent that range by a 4-bit word. The earliest microprocessors were 4-bit (word length) devices, and such 4-bit microprocessors are still widely used in such devices as toys, washing machines and domestic central heating controllers. They were followed by 8-bit microprocessors, e.g. the Motorola 6800, the Intel 8085A and the Zilog Z80. Now, 16-bit, 32-bit and 64-bit microprocessors are available; however, 8-bit microprocessors are still widely used for controllers.

### 2 *Address bus*

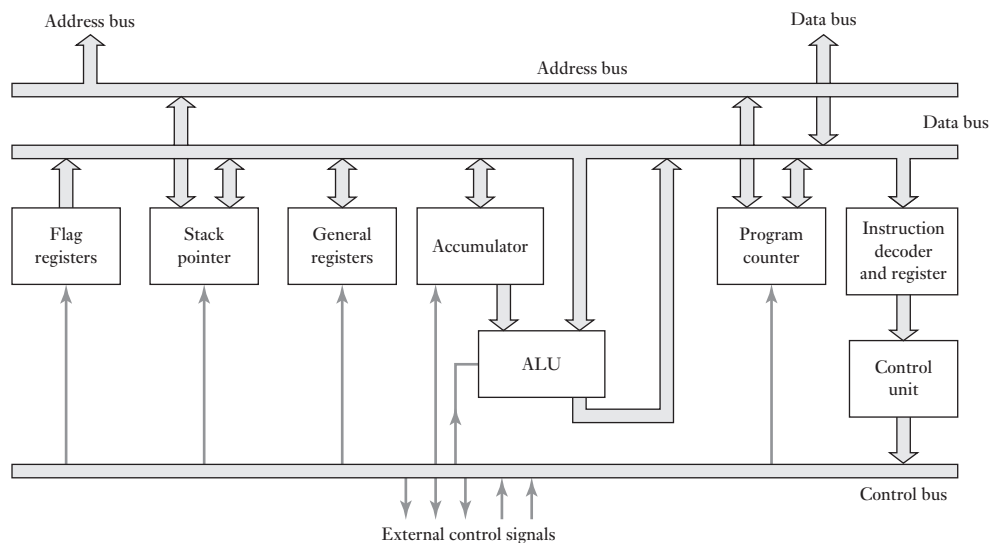
The **address bus** carries signals which indicate where data is to be found and so the selection of certain memory locations or input or output ports. Each storage location within a memory device has a unique identification, termed its address, so that the system is able to select a particular instruction or data item in the memory. Each input/output interface also has an address. When a particular address is selected by its address being placed on the address bus, only that location is open to the communications from the CPU. The CPU is thus able to communicate with just one location at a time. A computer with an 8-bit data bus has typically a 16-bit-wide address bus, i.e. 16 wires. This size of address bus enables  $2^{16}$  locations to be addressed;  $2^{16}$  is 65 536 locations and is usually written as 64K, where K is equal to 1024. The more memory that can be addressed, the greater the volume of data that can be stored and the larger and more sophisticated the programs that can be used.

### 3 *Control bus*

The signals relating to control actions are carried by the **control bus**. For example, it is necessary for the microprocessor to inform memory devices whether they are to read data from an input device or write data to an output device. The term READ is used for receiving a signal and WRITE for sending a signal. The control bus is also used to carry the system clock signals; these are to synchronise all the actions of the microprocessor system. The clock is a crystal-controlled oscillator and produces pulses at regular intervals.

## 10.2.2 The microprocessor

The microprocessor is generally referred to as the central processing unit (CPU). It is that part of the processor system which processes the data,



**Figure 10.2** General internal architecture of a microprocessor.

fetching instructions from memory, decoding them and executing them. The internal structure – the term **architecture** is used – depends on the microprocessor concerned. Figure 10.2 indicates, in a simplified manner, the general architecture of a microprocessor.

The following are the functions of the constituent parts of a microprocessor.

1 *Arithmetic and logic unit (ALU)*

The arithmetic and logic unit is responsible for performing the data manipulation.

2 *Registers*

Internal data that the CPU is currently using is temporarily held in a group of **registers** while instructions are being executed. These are memory locations within the microprocessor and are used to store information involved in program execution. A microprocessor will contain a group of registers, each type of register having a different function.

3 *Control unit*

The **control unit** determines the timing and sequence of operations. It generates the timing signals used to fetch a program instruction from memory and execute it. The Motorola 6800 uses a clock with a maximum frequency of 1 MHz, i.e. a clock period of 1  $\mu$ s, and instructions require between two and twelve clock cycles. Operations involving the microprocessor are reckoned in terms of the number of cycles they take.

There are a number of types of register, the number, size and types of registers varying from one microprocessor to another. The following are common types of registers.

1 *Accumulator register*

The accumulator register (A) is where data for an input to the arithmetic and logic unit is temporarily stored. In order for the CPU to be able to access, i.e. read, instructions or data in the memory, it has to supply the

address of the required memory word using the address bus. When this has been done, the required instructions or data can be read into the CPU using the data bus. Since only one memory location can be addressed at once, temporary storage has to be used when, for example, numbers are combined. For example, in the addition of two numbers, one of the numbers is fetched from one address and placed in the accumulator register while the CPU fetches the other number from the other memory address. Then the two numbers can be processed by the arithmetic and logic section of the CPU. The result is then transferred back into the accumulator register. The accumulator register is thus a temporary holding register for data to be operated on by the arithmetic and logic unit and also, after the operation, the register for holding the results. It is thus involved in all data transfers associated with the execution of arithmetic and logic operations.

2 *Status register, or condition code register or flag register*

This contains information concerning the result of the latest process carried out in the arithmetic and logic unit. It contains individual bits with each bit having special significance. The bits are called **flags**. The status of the latest operation is indicated by each flag with each flag being set or reset to indicate a specific status. For example, they can be used to indicate whether the last operation resulted in a negative result, a zero result, a carry output occurs (e.g. the sum of two binary numbers such as 1010 and 1100 is (1)0110 which might be bigger than the microprocessor's word size and thus there is a carry of a 1), an overflow occurs or the program is to be allowed to be interrupted to allow an external event to occur. The following are common flags:

| Flag | Set, i.e. 1          | Reset, i.e. 0                   |
|------|----------------------|---------------------------------|
| Z    | Result is zero       | Result is not zero              |
| N    | Result is negative   | Result is not negative          |
| C    | Carry is generated   | Carry is not generated          |
| V    | Overflow occurs      | Overflow does not occur         |
| I    | Interrupt is ignored | Interrupt is processed normally |

As an illustration, consider the state of the Z, N, C and V flags for the operation of adding the hex numbers 02 and 06. The result is 08. Since it is not zero, then Z is 0. The result is positive, so N is 0. There is no carry, so C is 0. The unsigned result is within the range  $-128$  to  $+127$  and so there is no overflow and V is 0. Now consider the flags when the hex numbers added are F9 and 08. The result is (1)01. The result is not zero, so Z is 0. Since it is positive, then N is 0. The unsigned result has a carry and so C is 1. The unsigned result is within the range  $-128$  to  $+127$  and so V is 0.

3 *Program counter register (PC) or instruction pointer (IP)*

This is the register used to allow the CPU to keep track of its position in a program. This register contains the address of the memory location that contains the next program instruction. As each instruction is executed, the program counter register is updated so that it contains the address of the memory location where the next instruction to be executed is stored. The program counter is incremented each time so that the CPU executes instructions sequentially unless an instruction, such as JUMP or BRANCH, changes the program counter out of that sequence.

#### 4 *Memory address register (MAR)*

This contains the address of data. Thus, for example, in the summing of two numbers the memory address register is loaded with the address of the first number. The data at the address is then moved to the accumulator. The memory address of the second number is then loaded into the memory address register. The data at this address is then added to the data in the accumulator. The result is then stored in a memory location addressed by the memory address register.

#### 5 *Instruction register (IR)*

This stores an instruction. After fetching an instruction from the memory via the data bus, the CPU stores it in the instruction register. After each such fetch, the microprocessor increments the program counter by one with the result that the program counter points to the next instruction waiting to be fetched. The instruction can then be decoded and used to execute an operation. This sequence is known as the **fetch–execute cycle**.

#### 6 *General-purpose registers*

These may serve as temporary storage for data or addresses and be used in operations involving transfers between other registers.

#### 7 *Stack pointer register (SP)*

The contents of this register form an address which defines the top of the stack in RAM. The **stack** is a special area of the memory in which program counter values can be stored when a subroutine part of a program is being used.

The number and form of the registers depends on the microprocessor concerned. For example, the Motorola 6800 microprocessor (Figure 10.3) has two accumulator registers, a status register, an index register, a stack pointer register and a program counter register. The status register has flag bits to show negative, zero, carry, overflow, half-carry and interrupt. The Motorola 6802 is similar but includes a small amount of RAM and a built-in clock generator.

The Intel 8085A microprocessor is a development of the earlier 8080 processor; the 8080 required an external clock generator, whereas the 8085A has an in-built clock generator. Programs written for the 8080 can be run on the 8085A. The 8085A has six general-purpose registers B, C, D, E, H and L, a stack pointer, a program counter, a flag register and two temporary registers. The general-purpose registers may be used as six 8-bit registers or in pairs BC, DE and HL as three 16-bit registers. Figure 10.4 shows a block diagram representation of the architecture.

As will be apparent from Figure 10.3 and Figure 10.4, microprocessors have a range of timing and control inputs and outputs. These provide outputs when a microprocessor is carrying out certain operations and inputs to influence control operations. In addition there are inputs related to interrupt controls. These are designed to allow program operation to be interrupted as a result of some external event.

### 10.2.3 Memory

The memory unit in a microprocessor system stores binary data and takes the form of one or more integrated circuits. The data may be program instruction codes or numbers being operated on.